

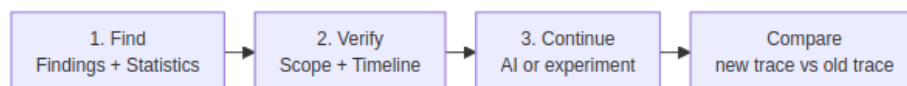
AI Assistant

BTFViewer's **AI Assistant** helps you investigate BTF trace data that BTFViewer has already measured. It does not replace Statistics or the timeline. Use it to organize evidence, test an explanation, find missing checks, and plan the next step.

Important: measured values come from BTFViewer. AI explanations are interpretations of that evidence. **What-if** and **Optimize** are estimates, not measured RTOS behavior.

Where to start

For most investigations, use this simple flow:



1. **Find the problem.** Start with **Analysis Findings** and the related Statistics section.
2. **Verify the evidence.** Use C1–Cn and the timeline to confirm that the event really happened in the expected region.
3. **Continue the investigation.** Ask AI to explain or challenge the evidence, or make one controlled change and capture another trace.
4. **Compare the result.** Use **Trace Compare** to measure whether the change helped.

AI is optional. You can complete an investigation with Statistics, the timeline, Investigation Notebook, and Trace Compare. The Notebook is the user-facing record; AI suggestions remain opt-in and do not edit it automatically.

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User guide

What the AI Assistant does

The AI Assistant receives structured information from BTFViewer. It does **not** read the full raw .btf file as free-form text.

It can use:

- **Analysis Findings** —deterministic clues generated by BTFViewer.
- **Statistics** —measured or derived metrics for the current scope.
- **Timeline evidence** —events and timestamps returned by BTFViewer tools.

- **Scope and Filters** —the active C1–Cn range and task/core/migration restrictions.
- **Trace Compare** —measured differences between two traces.
- **Investigation state** —the current finding, hypothesis, evidence, and checks.

It does not read firmware source code or ELF files.

What AI is good for Use AI when you want to:

- explain why a measured result may matter;
- connect several pieces of evidence;
- check whether an explanation is supported;
- look for an alternative explanation;
- identify missing evidence;
- plan the next measurement;
- summarize an investigation;
- compare two traces in plain language.

What AI should not decide by itself Do not treat these as measured facts unless BTFViewer provides supporting evidence:

- a root cause;
- a task dependency that is not visible in the trace;
- scheduler behavior that was not recorded;
- the effect of a proposed firmware change;
- a performance improvement predicted by What-if or Optimize.

Return to Statistics and the timeline whenever a conclusion matters.

Using the AI panel

The AI panel is designed around the current trace and investigation state.

Main controls Question box

Type a normal question about the current trace. The assistant receives the current BTFViewer context.

Start Investigation

Use this when you want BTFViewer to guide a full investigation. It is the best default for a new user.

The flow is:

Triage → Scope → Investigate → Verify → Experiment → Compare

The assistant should reuse evidence already collected instead of repeating the same checks.

Context

The collapsed Context line summarizes the current stage, scope, focus, context mode, and privacy state. Expand it when you need to check exactly what will be sent.

Important items are:

- active trace;
- full trace or C1–Cn scope;
- task/core/migration filters;
- selected task or finding;
- context mode;
- endpoint and privacy state.

Clear

Clears the AI conversation and current AI investigation state. It does not replace normal trace controls.

Language...

Sets the requested reply language.

Settings...

Opens AI connection, model, context, privacy, and related options.

Context modes BTFViewer provides three context sizes:

- **Compact** —lowest token use; best for focused questions.
- **Balanced** —default; enough context for most investigations.
- **Full Evidence** —sends more evidence; use when the smaller modes omit information needed for a difficult case.

A larger context does not automatically mean higher confidence. Confidence should come from evidence.

Built-in actions

The AI panel provides shortcuts for common tasks. The visible shortcuts are context-sensitive, and additional templates are available from **More templates**... Treat them as different questions that reuse the same investigation context, not as separate analysis systems.

Start Investigation **Use when:** you do not know which AI action to choose.

What it does: guides the investigation from the current evidence and chooses the next useful checks.

Expected result: a leading explanation, evidence, confidence/quality, and a next check.

Analysis Findings **Use when:** you want a short summary of the most important current Findings.

What it does: summarizes up to a few actionable Findings and points to the relevant Statistics or timeline check.

Does not: prove root cause.

Triage findings **Use when:** there are many Findings.

What it does: ranks the Findings that deserve attention first.

Does not: perform a full root-cause analysis.

Explain region **Use when:** C1–Cn surrounds an incident.

What it does: explains events inside the cursor range and keeps cited timestamps inside that range.

Investigate **Use when:** you already have a concrete performance problem.

What it does: gathers missing evidence, correlates related events, tests the leading explanation, and identifies an alternative.

Verify finding **Use when:** you want to check whether a selected Finding is actually supported.

What it does: checks the claim against available evidence and returns **Confirmed**, **Rejected**, or **Inconclusive**.

Explain finding **Use when:** a Finding is technically correct but difficult to understand.

What it does: explains the Finding and points to supporting evidence.

What-if **Use when:** evidence is strong enough to consider one possible change.

What it does: estimates the likely direction of change.

Important: the result is an estimate, not a scheduler simulation or measured result.

Optimize **Use when:** you want candidate improvement ideas after the problem has been verified.

What it does: ranks possible directions to test.

Important: validate every useful idea with a new trace.

Trace Compare **Use when:** two traces are open.

What it does: explains measured A/B differences from Trace Compare.

Choosing the right entry point

You can ask AI from several places. Each entry point should carry the evidence that is already visible there.

- **Analysis Findings → Ask AI:** use the selected Finding and its evidence.
- **Timeline event → Ask AI about this event:** use the selected task, core, and event time.
- **Timeline region → Explain this region with AI:** use C1–Cn.
- **Statistics → Query with AI…:** use the selected metric, task, and displayed samples.
- **Migration & Corridor Inspector → Investigate with AI:** use the selected migration path and current Inspector scope.
- **Trace Compare → Query with AI…:** use the comparison tables for the selected traces.
- **AI panel:** use the current trace, Scope, Filters, selection, and investigation state.

Scope matters Use **Limit to C1–Cn** when the problem belongs to one phase of the trace. AI evidence should then stay inside that interval unless it clearly states why information outside the range is needed.

A visual highlight is not a Filter. Task, core, and migration Filters affect the analysis context.

Investigation workflows

Use these workflows after selecting the task, Finding, or time window to investigate.

Investigation workflow

Step	Template or tool	Purpose
1	Triage findings / detect_anomalies	Rank the important Findings and open the related Statistics evidence
2	Investigate / investigate	Build hypotheses, alternatives, and an evidence plan

Step	Template or tool	Purpose
3	Verify finding	Decide whether the selected Finding is supported, rejected, or still inconclusive
4	<code>correlate_events + query_raw_metric</code>	Combine execution, blocking, migration, synchronization, and PI evidence for the selected task
5	<code>find_critical_path / detect_priority_inversion</code>	Inspect preemption, blocking, and priority-inversion paths when relevant
6	<code>build_task_dependency_graph / analyze_temporal_causality</code>	Organize dependency and happens-before relationships recorded by BTF
7	<code>rank_root_causes / challenge_conclusion</code>	Rank explanations and test credible alternatives
8	<code>find_related_findings / compare_tasks</code>	Check related Findings or compare suspect tasks
9	<code>set_cursors / zoom_to_range / highlight_task</code>	Bring the strongest evidence back to the timeline
10	Evidence & Validation	Review evidence, missing checks, contradictions, and the next useful check
11	<code>generate_report / close_investigation / export_investigation</code>	Record the result when the investigation is ready to close

The sequence is not a requirement to run every tool. The Investigation planner should reuse existing evidence and skip steps that do not apply.

The shortcuts are easier to understand when they are placed in a complete workflow.

First investigation Use this sequence when opening an unfamiliar trace:

1. Open **Analysis Findings** and choose one relevant warning or error.
2. Open the supporting Statistics section and confirm the measured symptom.
3. Select the affected task and inspect its timeline.
4. Put C1 and C2 around one representative incident. Add more cursors only when they help define a larger sequence.
5. Enable **Limit to C1–Cn** when the problem is specific to that interval.
6. Open the AI panel and use **Start Investigation**.
7. Follow the cited evidence instead of reading only the explanation.
8. Use **Verify finding** when the explanation appears plausible.
9. If evidence is incomplete, follow **Next check** or ask for the missing measurement.
10. Only after verification, use **What-if, Optimize**, or define your own controlled change.
11. Capture a new trace under comparable conditions.
12. Use **Trace Compare** to measure the result.

A useful investigation result is not a long AI answer. It is a short chain that can be checked:

Measured symptom
↓
Scoped incident
↓
Supporting event or metric
↓
Leading explanation
↓
Verification / contradiction
↓
Measured experiment result

Explain one event Use this when you already see a suspicious timeline segment.

1. Select the segment.
2. Choose **Ask AI about this event**.
3. Check the task, core, and timestamp included in the request.
4. Follow any jump:TIME reference back to the timeline.
5. If the event is part of a longer sequence, place C1–Cn around the sequence and continue with **Explain region**.

This path is intended for a local question. It should not automatically become a full-trace root-cause analysis.

Explain one region Use this when the important behavior is contained between cursors.

1. Place at least two cursors.
2. Confirm the intended interval.
3. Enable **Limit to C1–Cn** if Statistics and AI should use only that range.
4. Use **Explain this region with AI** or **Explain region**.
5. Check that cited timestamps belong to the intended interval.
6. Expand the range only when evidence shows that an earlier cause or later effect must be inspected.

Verify a Finding A Finding is a clue produced by deterministic rules or statistics. It is not automatically a cause.

A useful verification should answer:

- What exactly is the claim?
- Which measured values support it?
- Which timeline events support it?
- Is the evidence inside the current Scope?
- Is there contradictory evidence?
- Is another explanation still plausible?
- What evidence is still missing?

The final status should be understandable without reading internal tool logs.

What-if and Optimize workflow `what_if` and `optimize_experiment` are **heuristic slice-replay** tools. They reuse measured execution slices and estimate the effect of changes such as task placement, migration reduction, blocking reduction, or priority adjustment.

They do **not** simulate an RTOS kernel or a deterministic scheduler. Their output is an estimate used to choose an experiment, not proof that the change will work.

Goal	Use	Example
Test one concrete idea	What-if → what_ if	Pin a task to one core, raise a priority, reduce contention
Rank several ideas	Optimize → optimize_ experiment	Compare placement, contention, priority, or migration changes
Get qualitative advice	optimize	Explain likely mitigations without a scored simulation
Decide what to measure next	recommend_ experiments	Turn the estimate into a real validation experiment

Read the result by comparing the measured **baseline** with the **simulated** estimate. Depending on the case, useful fields include migrations, blocking time, load-balance score, and experiment cost.

Medium confidence means the idea may be worth testing on the real system. **Low confidence** usually means the requested change is vague or the trace contains too little evidence.

Always finish the workflow with a real measurement:

```
Verified evidence
  ↓
What-if / Optimize estimate
  ↓
One controlled system change
  ↓
Capture a new trace
  ↓
Trace Compare
```

The new trace, not the estimate, determines whether the change worked.

Common use cases

Long response-time tail Start with the response-time distribution and p95/p99/Max. Scope one slow instance, then inspect execution, dispatch, blocking, and nearby

preemption. AI can help organize these components, but the cited values should remain traceable to Statistics.

Blocking or synchronization problem Start from blocking or mutex-related Statistics. Scope one incident and inspect the owner/waiter sequence that is actually visible in the trace. Treat inferred handoff or waiter relationships as heuristic when the trace format does not contain the kernel's real wait queue.

Core migration or ping-pong Use Migration & Corridor Inspector first. Check migration count, dwell, ping-pong behavior, load balance, and the selected path. AI can correlate the path with nearby scheduling or synchronization evidence. A migration by itself is not proof of a performance problem.

Suspected priority inversion Use the relevant priority, blocking, mutex, and scheduling evidence. AI may use `detect_priority_inversion`, but the conclusion still depends on what the trace recorded. Missing kernel state must not be invented.

Regression after a change Open comparable baseline and candidate traces. Use Trace Compare before asking AI. AI should explain the measured differences, localize the likely task/region, and identify the next check. The comparison result is stronger evidence than an estimated improvement.

Periodic task jitter Start with period, dispatch, execution, and response distributions. Use C1–Cn around a representative outlier. Check whether the disturbance repeats and whether it aligns with blocking, preemption, migration, or another workload phase.

Worked example

Suppose Analysis Findings reports unusually high response-time p99 for Worker. A useful investigation is:

1. Open the response-time Statistics section and confirm that p99 is actually high.
2. Open the distribution and inspect one representative slow sample.
3. Use **Show on timeline** or the related timestamp to reach that event.
4. Put C1 and C2 around the incident.
5. Enable **Limit to C1-Cn**.
6. Ask AI to investigate the selected Finding.
7. Check whether the response delay is associated with execution, dispatch, blocking, preemption, migration, or another measured component.
8. Use **Verify finding** to test the leading explanation and look for contradicting evidence.
9. If the result remains inconclusive, follow the missing-evidence check rather than forcing a root-cause statement.
10. If a concrete change is justified, capture another trace and compare the same metric.

A good final note is concise:

Observation:

Worker response-time p99 is elevated in the selected phase.

Evidence:

The slow sample is inside C1-C2 and coincides with the measured
↪ scheduling/blocking evidence cited by BTFViewer.

Conclusion:

Leading explanation supported / rejected / still inconclusive.

Next action:

One measurable change or one additional evidence check.

Continue the investigation

The AI conversation should continue from the current investigation instead of restarting after every reply.

Use **Next check** when the Evidence panel already identifies the most useful missing check. Use a normal follow-up question when you want to change the question, challenge the explanation, or ask for clarification.

When BTFViewer exposes a runnable next step, the action should continue with the current investigation context. Completed evidence should be reused. Repeating the same read-only queries is useful only when Scope, Filters, trace selection, or required evidence has changed.

A sensible stopping condition is reached when:

- the main claim has enough evidence for the intended engineering decision;
- a credible alternative has been checked;
- remaining missing evidence is unlikely to change the decision; or
- the trace does not contain enough information and a new capture is required.

“Inconclusive” is a valid result.

Understanding the result

Evidence & Validation This panel is the quickest place to decide whether the AI answer is useful.

Read it in this order:

1. **Verdict** —current result of the investigation.
2. **Leading explanation** —the best explanation supported so far.
3. **Direct / Timeline evidence** —measured evidence that can be checked in BTFViewer.
4. **Checks** —claims that were tested.
5. **Alternatives** —other explanations considered.
6. **Missing evidence** —information still needed.
7. **Next check** —the most useful next action.
8. **Investigation details** —deeper diagnostic information such as tools used and confidence evolution.

Do not start with the deepest details. For normal use, the first seven items are enough.

Evidence strength BTFViewer may distinguish evidence as measured, derived, heuristic, configured, or simulated/estimated. These are not equally strong.

Prefer this order when making a decision:

measured event/value
↓
derived statistic
↓
heuristic interpretation
↓
estimate / simulation

Confidence Confidence describes how well the available evidence supports the explanation. It is not the probability that the AI is correct.

A high-confidence causal statement should have trace evidence and verification, not only a plausible narrative.

Apply, Skip, and Undo Read-only evidence queries can run immediately.

Actions that change the viewer may appear as action cards:

- **Navigation** —move or zoom the timeline.
- **Scope** —change the active analysis range.
- **Filter** —change task/core/migration restrictions.
- **Annotation** —add marks or notes.
- **Export** —save a report or investigation package.
- **Calculation** —run an evidence or estimate operation.

Use **Apply** to accept a pending viewer change and **Skip** to leave the viewer unchanged. **Undo** restores supported viewer state such as zoom, cursors, highlight, Scope, and Filters.

AI tools

Most users do not need to call tools by name. The AI Assistant chooses them when needed. This section explains what each group is for.

1. Navigate and show evidence These tools change what you see in the viewer.

- `set_cursors` —place C1–Cn at evidence timestamps.

- `zoom_to_range` —zoom to a time interval.
- `highlight_task` —highlight one task.
- `set_view_mode` —switch Task/Core or orientation.
- `open_corridor_inspector` —open Migration & Corridor Inspector.
- `open_statistics_section` —open a Statistics section without changing Scope.
- `add_annotation` —add a note at a timestamp.
- `bookmark_finding` —preserve a Finding as a bookmark.
- `clear_marks` —clear selected marks/cursors/bookmarks.
- `reset_view` —return to the full trace view.

2. Measure and search These are read-only evidence tools.

- `query_raw_metric` —read scoped task metrics such as execution, blocking, migration, synchronization, priority inheritance, activation, and ready gaps.
- `search_timeline` —find matching timeline events and timestamps.
- `analyze_distribution` —calculate percentiles, spread, CV, and outlier information.
- `analyze_periodicity` —inspect expected period and timing jitter.
- `check_budget` —compare timing against a configured budget or deadline.
- `decompose_response_time` —show relative response-delay components.

3. Find and connect related problems

- `detect_anomalies` —rank current Analysis Findings.
- `cluster_findings` —group related Findings.
- `cluster_incidents` —group events that occur close together.
- `investigate` —build an investigation graph for a Finding.
- `plan_investigation` —choose useful next checks.
- `suggest_scope` —suggest a focused time range.
- `correlate_events` —combine execution, blocking, migration, synchronization, priority, and search hits around an event.
- `find_related_findings` —locate Findings related to the current issue.
- `find_critical_path` —inspect blocking, preemption, and mutex activity around an incident.
- `build_task_dependency_graph` —show observed wait/preempt/migrate/priority-inheritance relationships.
- `analyze_temporal_causality` —build an observed happens-before sequence.

- `build_causal_chain` —organize causal, correlated, and temporal links without silently treating correlation as causation.
- `rank_root_causes` —rank supported explanations.

4. Verify and challenge an explanation

- `verify_claim` —classify a claim as supported, partial, or unsupported.
- `detect_contradictions` —search for evidence against the current explanation.
- `challenge_conclusion` —propose credible alternatives and missing evidence.
- `assess_evidence_sufficiency` —decide whether to stop, continue, or revise the hypothesis.
- `manage_hypotheses` —track hypothesis state.
- `detect_priority_inversion` —inspect evidence consistent with priority inversion.
- `explain_finding` —explain a selected Finding.

5. Compare traces and tasks

- `trigger_compare` —obtain/open Trace Compare for two traces.
- `compare_performance` —return structured A/B metric differences.
- `regression_explain` —explain the primary measured regression.
- `regression_localize` —locate a likely task and region associated with the regression.
- `compare_tasks` —compare two tasks.
- `baseline_score` —compare current behavior with a stored baseline.
- `analyze_traces` —compare scheduling behavior across several open traces.
- `generate_fingerprint` —summarize scheduling, synchronization, and timing bands.

6. Plan and validate an experiment

 These tools come after evidence gathering.

- `what_if` —estimate one proposed change.
- `optimize` —suggest qualitative improvement directions.
- `optimize_experiment` —rank candidate changes to test.
- `recommend_experiments` —suggest validation experiments.
- `generate_experiment_plan` —create a concrete firmware/bench test plan.
- `validate_experiment` —compare a prediction with the measured result.

- `record_experiment_outcome` —save the measured outcome for later matching.
- `find_similar_investigations` —find recorded cases with a similar fingerprint.

7. Report and preserve the investigation

- `generate_report` —generate structured engineering text.
- `export_report` —save a diagnostic report.
- `export_investigation` —save the full investigation package.
- `summarize_investigation_context` —create a compact investigation summary.
- `investigation_memory` —store or recall similar investigation records.
- `close_investigation` —close the current investigation with its conclusion and confidence.

Supporting host tool

- `interpret_query` —interprets a free-form question and prepares the appropriate investigation request before the main run.

Configuration and privacy

Connect an endpoint Open **Settings** → **AI** and configure the provider, endpoint, model, and authentication required by that provider. Use **Test connection** before starting an investigation.

BTFViewer can test model listing, chat, structured output, and tool calling when supported by the endpoint.

Import provider settings **Settings** → **AI** → **Import**... loads preset, base URL, model, API key, and authentication mode from a JSON file (see `examples/ai/*.json`). A preset name that isn't already in the list—for example `openrouter` or `deepseek`—is added as a new preset instead of being folded into Custom. Each imported preset keeps its own Base URL, Model, API key, authentication mode, and TLS setting, independent from the built-in presets and from every other imported preset.

Import updates the Settings dialog immediately, but nothing is written to disk (btf_viewer.rc) or browser storage until you click **OK**. **Cancel** discards the import.

Importing the same preset again —after normalizing case and punctuation, so Open-Router and open-router match the same entry —updates that preset's saved values instead of creating a duplicate. A file that lists multiple presets adds all of them, without removing presets you imported earlier.

Choose a model For BTFViewer, tool reliability matters more than fluent prose. Prefer a model that can:

- follow structured instructions;
- call tools reliably;
- preserve tool-call state across follow-up turns;
- handle the selected context size;
- distinguish evidence from inference.

Use the project's benchmark results when choosing between supported models. Model recommendations change over time and should not be inferred from this document alone.

Privacy The AI panel indicates whether the configured endpoint is local or cloud-based. Review the current privacy state before sending trace-derived information to a remote service.

Cloud privacy handling may sanitize annotations and optionally alias task names. Sensitive configurations can block cloud sending.

Credentials Use the credential method provided by the current BTFViewer implementation and provider. Do not place API keys in trace files, reports, screenshots, or shared investigation packages.

Reset to Defaults **Settings → Reset to Defaults** restores every tab in the dialog to its built-in default, including layout, Statistics presentation, and AI settings. For AI specifically, this means:

- every preset added by Import...is removed;

- the preset list returns to the four built-ins (Custom, Ollama, OpenAI, Gemini);
- the current AI investigation/conversation is cleared;
- saved API keys and tokens are cleared, for the built-in presets and any imported ones.

Like Import, Reset to Defaults only changes the dialog until you click **OK**. Reopening Settings and clicking **Cancel** afterward leaves your previous configuration —presets, credentials, and conversation —untouched.

Viewer behavior

A few rules are important when interpreting AI actions:

- Read-only evidence queries do not change the timeline.
 - Navigation may move the viewport, cursors, or highlight.
 - Scope and Filter changes affect later Statistics and AI evidence.
 - A highlight is visual only; it is not a Filter.
 - `open_statistics_section` opens the requested Statistics section without changing Scope.
 - Trace Compare requires at least two loaded traces.
 - What-if and Optimize results must remain labeled as estimates.
 - Saved reports should preserve the evidence hierarchy while avoiding unnecessary internal tool chatter.
-

Troubleshooting

AI is unavailable Check **Settings → AI**, the configured endpoint, model, authentication, and **Test connection**.

The answer is too broad Narrow the problem first:

1. select the relevant task;
2. place C1–Cn around the incident;
3. enable **Limit to C1–Cn** if appropriate;
4. ask one specific question.

The answer cites the wrong region Confirm the current Scope and Filters. If C1–Cn is active, verify that cited jump:TIME evidence lies inside the range.

The answer sounds plausible but has weak evidence Use **Verify finding** or ask the assistant to challenge the conclusion. Then check the cited Statistics and timeline events yourself.

The model repeats tool calls Use a more focused Scope, clear an excessively long conversation, or use a model with more reliable tool calling. A verification action should reuse existing evidence whenever possible.

A local model's reply or proposal is cut off If **Gather evidence with AI** (or another Notebook collaboration) returns an incomplete answer –a `btf-viewer-nb-proposal` JSON block that stops mid-way –the model hit its output limit or stopped early while generating. The Notebook shows a hint in place of the truncated text.

BTFViewer already asks for a proposal-sized reply budget (4096 tokens) on every Notebook collaboration, so the remaining causes are the model and the server:

1. Use a larger or instruction-tuned model; small models lose track of a long structured reply and emit an end-of-text token early.
2. Reduce the request size: narrow the Scope, trim the investigation to the essential evidence before collaborating, and clear a long chat.
3. Give a local server a large input window and restart it. For Ollama, `OLLAMA_CONTEXT_LENGTH=8192 ollama serve` (or larger), or add `PARAMETER num_ctx 8192` to a Modelfile and `ollama create`. A value set only for an interactive `ollama run` session does not apply to API calls.

`num_ctx` (input window) is separate from the reply budget. **Compact** context mode is fine for Notebook collaboration –the 4096-token proposal budget overrides its usual 500-token reply cap.

Connection errors For browser CORS, authentication, TLS, model-not-found, timeout, or provider-specific errors, first use **Test connection**. If the endpoint itself does not respond correctly, fix the endpoint before debugging the investigation workflow.

Engineering reference

Investigation Case

BTFViewer keeps one shared **Investigation Case** (btf-investigation-case) so that several AI actions can continue the same investigation instead of rebuilding context from the conversation each time.

The case can contain:

- the current question;
- Scope: trace, C1–Cn, tasks, and cores;
- observations;
- hypotheses and their state:
 - supported
 - possible
 - need evidence
 - rejected
- supporting and contradicting evidence;
- Evidence Graph;
- Coverage;
- falsification checks;
- missing evidence;
- conclusion;
- validation;
- experiment prediction and measured outcome.

The Investigation Case is a continuity mechanism. It does **not** make an AI conclusion true. The important part is still the evidence that can be checked in BTFViewer.

Validation After a final assistant reply, the host-side validator checks claims that can be verified mechanically. It extracts references such as:

- jump:TIME
- Task[id]

and can flag:

- invented task names;
- timestamps that do not exist;

- timestamps outside the active cursor range.

This protects the investigation from claims that sound plausible but cannot be mapped back to the trace.

Model capability check **Test connection** can append a model-capability result covering supported features such as:

- live chat;
- structured output;
- tool calling.

These checks describe endpoint capability. They are separate from the benchmark suite, which measures investigation quality on known cases.

Headless evaluation The same investigation behavior can be checked without the GUI:

```
make -C BTFViewer ai-test
```

or:

```
python builds/btf_viewer.py ai-test \  
  --dataset tests/ai \  
  --fail-under 70
```

Investigation modes Host-side modes such as:

```
quick  
diagnose  
compare  
optimize  
report
```

still map to existing investigation templates internally. They are not intended to become a permanent row of additional UI buttons. The same workflows should remain available through the dynamic shortcuts or **More templates**...

Regardless of the mode, important conclusions should still be verified against Statistics and the timeline.

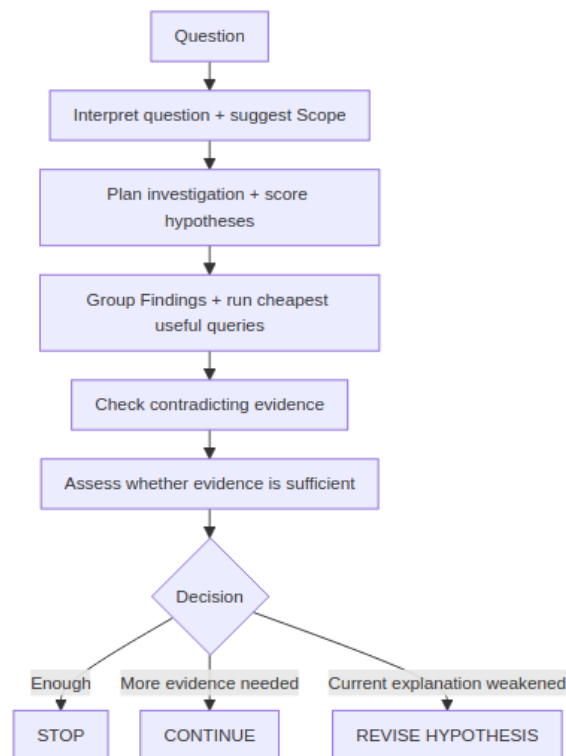
Investigation planner

The Investigation planner is host-side logic that decides **what evidence should be checked next**.

Its main rule is:

Cheapest useful evidence first.

The planner should prefer low-cost checks that can change the conclusion before requesting broader or more expensive evidence. It should also avoid repeating work that the current Investigation Case already contains.



Planner flow The planner normally proceeds in this order:

1. Interpret the question.
2. Suggest or reuse an appropriate Scope.
3. Build or rank hypotheses.
4. Group related Findings when useful.
5. Run the least expensive evidence checks first.

6. Search for contradictory evidence.
7. Judge whether the collected evidence is sufficient.
8. Stop, continue, or revise the current hypothesis.

This is intended to prevent a full investigation from becoming a fixed sequence of every available tool.

Planner tools and helpers

Tool / helper	Purpose	Result
plan_investigation	Rank hypotheses and choose a low-cost evidence sequence from the question and Findings	Ordered investigation plan
suggest_scope	Suggest task, related tasks, evidence timestamps, or reuse current cursors	Candidate analysis Scope
detect_contradictions	Test whether available evidence weakens the current explanation	SUPPORTED, CONTRADICTED, or INSUFFICIENT
assess_evidence_sufficiency	Judge whether current evidence is enough	Stop, continue, or revise
score_hypotheses	Compute evidence-weighted hypothesis scores; host helper, not a GUI tool	Ranked hypotheses
cluster_findings	Group Findings by shared task or pattern	Related Finding groups
generate_fingerprint	Summarize scheduling, synchronization, and timing bands	HIGH / MEDIUM / LOW feature bands
find_similar_investigations	Compare the current fingerprint with recorded experiment outcomes	Similar recorded cases

Tool / helper	Purpose	Result
<code>regression_localize</code>	Use A/B differences to locate the likely task, region, and mechanism	Regression focus
<code>build_causal_chain</code>	Organize causal, correlated, and temporal edges	Evidence relationship chain
<code>generate_experiment_plan</code>	Rank pinning, contention, priority, or related experiments	Candidate experiments
<code>record_experiment_outcome</code>	Save the measured experiment result	Reusable investigation record
<code>score_investigation_metrics</code>	Host-side close-out scoring; not model-facing	Investigation quality metrics

`build_causal_chain` must keep causal, correlated, and temporal edges distinct. A disclaimer is required when the evidence does not support a true causal claim.

Investigation quality metrics When an investigation is closed, host-side scoring can include:

- `evidence_efficiency`
- `investigation_cost`
- `false_confidence`
- `falsification_quality`
- `scope_accuracy`
- `stop_efficiency`

These metrics can also contribute to benchmark-case scoring, including adversarial-case rates.

Stop conditions The planner should stop when:

- the current explanation has enough evidence for the intended decision;
- a credible alternative has been checked;
- additional evidence is unlikely to change the result; or

- the current trace does not contain the information needed to continue.

If the evidence weakens the current explanation, the planner should **revise the hypothesis**, not simply collect more supporting evidence.

UI rule Do **not** solve planner limitations by adding more chat templates or more primary buttons. The planner is intended to improve the depth and efficiency of the existing investigation workflow.

Saved results and reports

The AI conversation can be useful during exploration, but the engineering result should be preserved separately.

A useful saved investigation contains:

- trace identity;
- Scope and Filters;
- observation;
- supporting evidence references;
- contradicting or missing evidence;
- current conclusion;
- confidence/evidence quality;
- experiment or next action.

Use the Investigation Notebook for concise investigation records. Use exported AI reports when a shareable narrative is needed. Avoid making raw tool logs the main report.

Engine limits

Some tools summarize or infer relationships from BTF evidence. Their names should not be interpreted as capabilities beyond the recorded trace.

Engine	What it provides	Important limit
<code>analyze_temporal_causality</code>	Happens-before relationships from recorded evidence and <code>jump:TIME</code>	Not kernel event replay
<code>build_task_dependency_graph</code>	BTF synchronization, preemption, migration, and PI edges around tasks	Not a complete ISR or kernel-object graph
<code>decompose_response_time</code>	Relative contribution derived from available Finding magnitudes	Not cycle-accurate response-time reconstruction
<code>rank_root_causes</code>	Ranking of hypotheses or Finding groups	The rank is not a probability
<code>investigation_memory</code>	Local investigation storage and recall	Not a shared team knowledge base
<code>cluster_incidents</code>	Groups incidents by time proximity	Does not prove a shared cause
<code>analyze_distribution</code>	Distribution analysis from available BTF samples	Cannot create a series the trace/parser does not provide
<code>analyze_periodicity</code>	Inter-arrival period and jitter analysis	Not a kernel timer model
<code>simulate_schedule</code>	Internal helper used by What-if estimation	Not an RTOS scheduler simulator or user-facing tool

BTFViewer does not reconstruct unrecorded kernel state, inspect ELF/source code, or perform hardware-aware scheduler simulation. If the trace does not contain enough evidence, the correct result is **insufficient evidence**.

Complete tool reference

This section is intended for implementation and debugging. Normal users can stop at [AI tools](#).

Tool	Tool	Tool
add_annotation	analyze_distribution	analyze_periodicity
analyze_temporal_causality	analyze_traces	assess_evidence_sufficiency
baseline_score	bookmark_finding	build_causal_chain
build_task_dependency_graph	challenge_conclusion	check_budget
clear_marks	close_investigation	cluster_findings
cluster_incidents	compare_performance	compare_tasks
correlate_events	decompose_response_time	detect_anomalies
detect_contradictions	detect_priority_inversion	explain_finding
export_investigation	export_report	find_critical_path
find_related_findings	find_similar_investigations	generate_experiment_plan
generate_fingerprint	generate_report	highlight_task
interpret_query	investigate	investigation_memory
manage_hypotheses	open_corridor_inspector	open_statistics_section
optimize	optimize_experiment	plan_investigation
query_raw_metric	rank_root_causes	recommend_experiments
record_experiment_outcome	regression_explain	regression_localize
reset_view	search_timeline	set_cursors

Tool	Tool	Tool
set_view_mode	suggest_scope	summarize_investigation_context
trigger_compare	validate_experiment	verify_claim
what_if	zoom_to_range	

Models without native tool calling may use BTFViewer's fallback tool-call format. For investigation-heavy work, a model with reliable native tool calling is preferred.

CLI and regression checks

The Desktop CLI includes `ai-test` for AI evidence and validator regression testing. Offline fixtures are used by default; `--models` runs configured live endpoints. CLI behavior is Desktop-only and does not need a matching Web command.

Use CLI regression checks to answer questions such as:

- does a known trace still produce the expected Findings?
- does a comparison still classify the same regression?
- does a model/tool change break the investigation workflow?
- does a saved investigation remain reproducible?

Do not treat `btf_viewer.html` alone as proof that every CLI command is available.

Benchmark results

BTFViewer's benchmark suite checks whether an AI model can investigate known BTF trace problems reliably. It is designed for **BTFViewer use**, not as a general-purpose LLM ranking.

The recorded results below are from **2026-09-04** using **17 test cases**. Unless noted otherwise, the model table uses **Full evidence**.

What the score means The benchmark evaluates the parts of an investigation that matter most in BTFViewer:

Area	What is checked
Finding	Does the model identify the expected problem?
Evidence	Are cited tasks, metrics, events, and timestamps supported by the trace?
Root cause	Does the explanation match the expected diagnosis rather than a misleading clue?
Calibration	Does confidence match the available evidence?
Safety	Does the model avoid invented or out-of-scope claims?

The **Overall** value is a weighted engineering score. It is **not a probability** that the answer is correct. In the recorded suite, Overall ≥ 70 is treated as PASS.

The suite also includes adversarial cases. These contain plausible but misleading clues, such as mutex activity near a CPU-starvation problem or correlation that does not establish causation. They test whether the model follows evidence instead of simply accepting the most obvious explanation.

Recorded model results

Model	Use	Overall	Passed	Mean time / case	Main observation
qwen3.5:9b	Local / practical	88	14/17	16.2 s	Best practical local reference; strong Evidence and Root-cause scores
qwen3.8:27b	Local / high latency	86	13/17	332 s	Similar score, but much slower in the recorded run
gemini-3.5-flash-lite	Cloud / fast	86	14/17	3.0 s	Fastest recorded shipped cloud reference

Model	Use	Overall	Passed	Mean time / case	Main observation
gemini-3.7-flash	Cloud	86	15/17	6.1 s	Highest pass count among the shipped Gemini references
gemini-3.8-flash	Cloud	85	14/17	9.1 s	Good with Balanced/Full evidence; weaker recorded Compact result

More detailed component scores:

Model	Finding	Evidence	Root cause	Calibration
qwen3.5:9b	85	93	82	80
qwen3.8:27b	88	94	65	80
gemini-3.5-flash-lite	82	94	65	80
gemini-3.7-flash	91	91	59	80
gemini-3.8-flash	88	91	59	80

Two additional cloud models were recorded from private configurations. They are useful as references but are **not part of the shipped benchmark model set**:

Model	Overall	Passed	Mean time / case
gpt-5.6-sol	88	15/17	9.6 s
claude-sonnet-5	82	12/17	15.9 s

How to read the results The results show that a higher model size or a larger context does not automatically produce a better BTFViewer investigation.

For local use, qwen3.5:9b is the practical reference in this test set. It reached Overall 88 with much lower recorded latency than qwen3.8:27b. The larger local model therefore did not provide enough measured benefit to offset its much higher runtime in this benchmark.

For cloud use, the Gemini models are close in overall score, but their behavior differs. gemini-3.5-flash-lite is the fastest recorded option, while gemini-3.7-flash completed the most cases successfully. This makes latency and investigation success more useful selection criteria than Overall alone.

Root-cause score deserves special attention. Several models identify the right Finding and cite valid evidence but score lower when deciding what actually caused the problem. BTFViewer therefore keeps verification and contradiction checks separate from the first explanation.

Context mode results BTFViewer can provide three levels of AI context:

Mode	What the model receives	Best fit
Compact	Smallest useful context	Quick questions and lower token use
Balanced	More supporting context without sending everything	Normal investigation
Full evidence	Largest evidence package	Difficult or ambiguous cases

The recorded benchmark shows:

- Compact used fewer tokens for every tested model.
- Fewer tokens did **not** always mean lower latency.
- More context did **not** always improve the result.
- qwen3.5:9b recorded its highest local pass count in Balanced mode.
- gemini-3.8-flash performed noticeably worse in Compact than in Balanced or Full evidence.

The practical conclusion is to use **Balanced** for normal investigation and move to **Full evidence** when the case needs broader evidence. Compact is useful when token use matters and the question is narrow.

What the test cases cover The 17-case suite includes normal and deliberately misleading trace problems. Major areas include:

- migration and load imbalance;
- mutex contention and priority inversion;
- deadline and response-time problems;
- period and jitter analysis;
- preemption;
- trace regression;
- explaining a selected region;
- waiter/owner handoff interpretation;
- choosing the correct Statistics page;
- distinguishing correlation from causation;
- rejecting timestamps outside the active Scope.

A benchmark case defines expected facts and evidence rather than requiring one exact wording. This allows different models to explain the same correct diagnosis in different ways.

Why evidence validation matters BTFViewer checks more than whether the answer sounds reasonable. The benchmark can reject behaviors such as:

- inventing task names or metric values;
- citing a `jump:TIME` that is not present or is outside `C1–Cn`;
- treating an estimated What-if result as a measured result;
- claiming causation when the trace only supports correlation;
- confirming a misleading clue before gathering the required evidence.

This is why a model with a strong general answer can still receive a lower BTFViewer benchmark score.

Using the results Use the benchmark as a **model-selection guide**, not as a permanent ranking.

For a new model or endpoint, compare:

1. investigation pass rate;
2. Evidence and Root-cause scores;
3. latency;
4. token use;

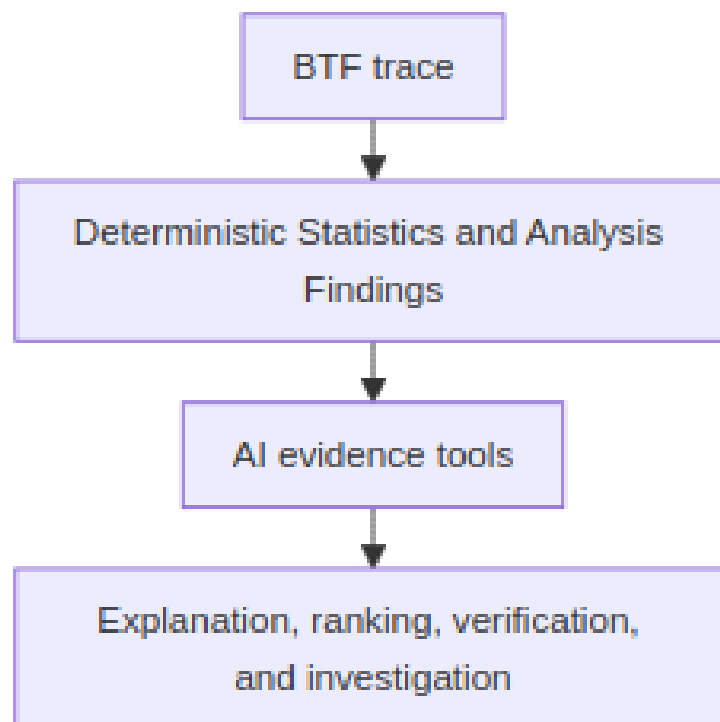
5. for local models, memory use.

The benchmark can be rerun with the BTFViewer ai-test workflow and the detailed report is written to [AI_BENCHMARK.md](#). The implementation details of the benchmark runner are intentionally kept out of this guide.

Implementation notes

Analysis vs AI tools Measured facts should come from BTFViewer's deterministic Statistics and Analysis Findings first. AI is responsible for **organizing, explaining, ranking, correlating, and challenging** those facts.

Do not add an AI tool when a Statistics page already provides the measurement. For example, AI does not need a second timeline-anomaly detector, another histogram engine, or a separate jitter calculator just to reproduce existing Statistics.



The responsibility boundary is:

- **Statistics / Analysis Findings:** measure and identify evidence.
- **AI tools:** connect evidence, test explanations, and guide the next check.

- **Timeline:** verify the actual event.
- **What-if / Optimize:** estimate experiments only.
- **Trace Compare:** measure the result after a real change.

AI must not invent kernel response time, inspect source/ELF data that BTFViewer did not provide, or claim to simulate the real scheduler.

Keep measured evidence separate from AI interpretation BTFViewer Statistics, Findings, timeline events, and Trace Compare are the evidence layer. AI organizes and tests that evidence.

Reuse evidence A later verification step should reuse evidence already collected in the same investigation. It should query again only when information is missing, stale, or outside the required Scope.

Do not hide state changes Navigation, Scope, Filter, and annotation changes should remain visible to the user and reversible where supported.

Avoid premature root-cause language Use **Leading explanation** until the evidence chain has been verified. Use **Root cause** only when the available trace evidence supports that stronger statement.

Keep the interface usable without AI Statistics, timeline verification, Investigation Notebook, and Trace Compare must remain sufficient for a complete manual investigation.

Documentation navigation

- **README.md** —basic BTFViewer operation.
- **WORKFLOWS.md** —step-by-step investigation workflow.
- **STATISTICS.md** —meaning of each measurement.
- **AI.md** —AI-assisted investigation and tool behavior.